

MEGN 441

Introduction to Robotics

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HW 1 due 09/01

DC Motors

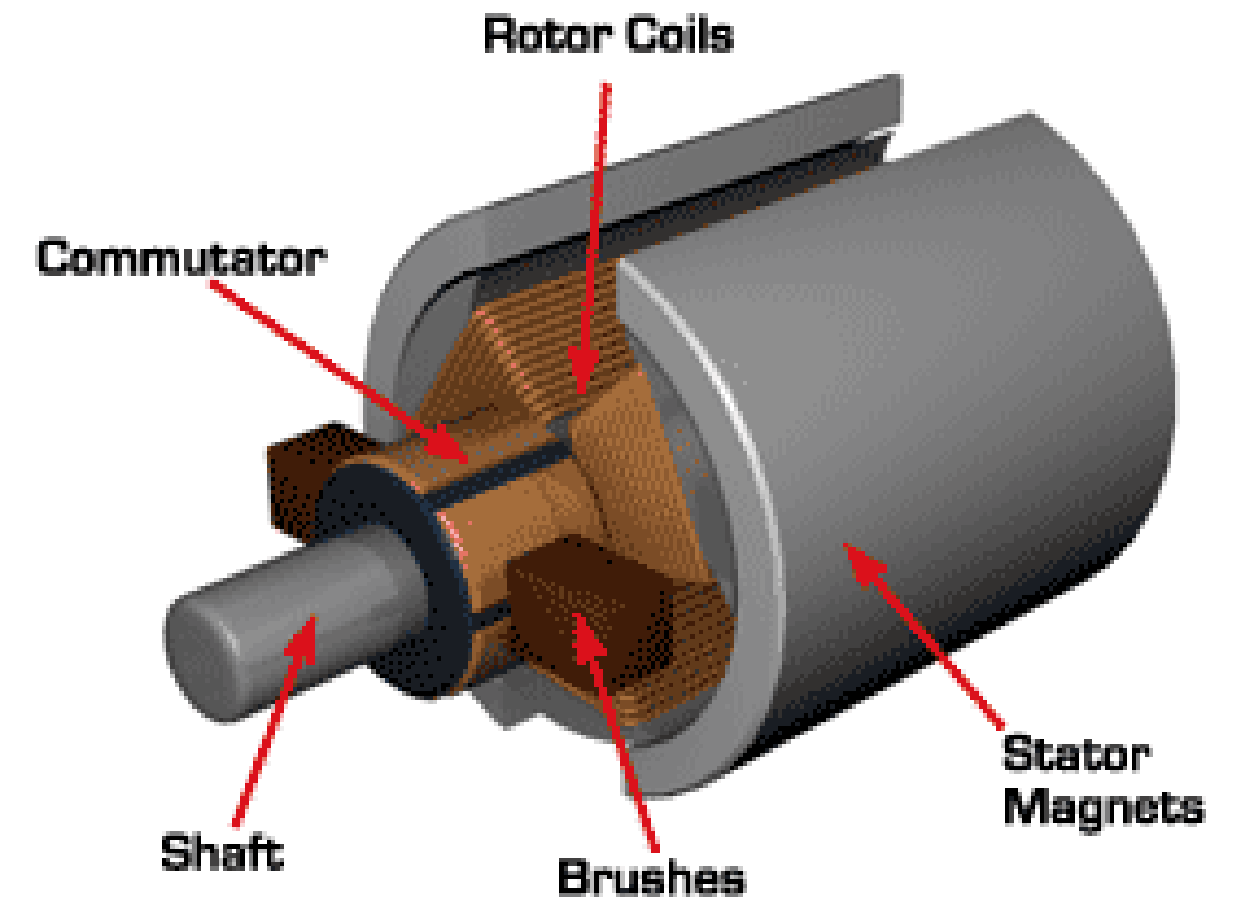
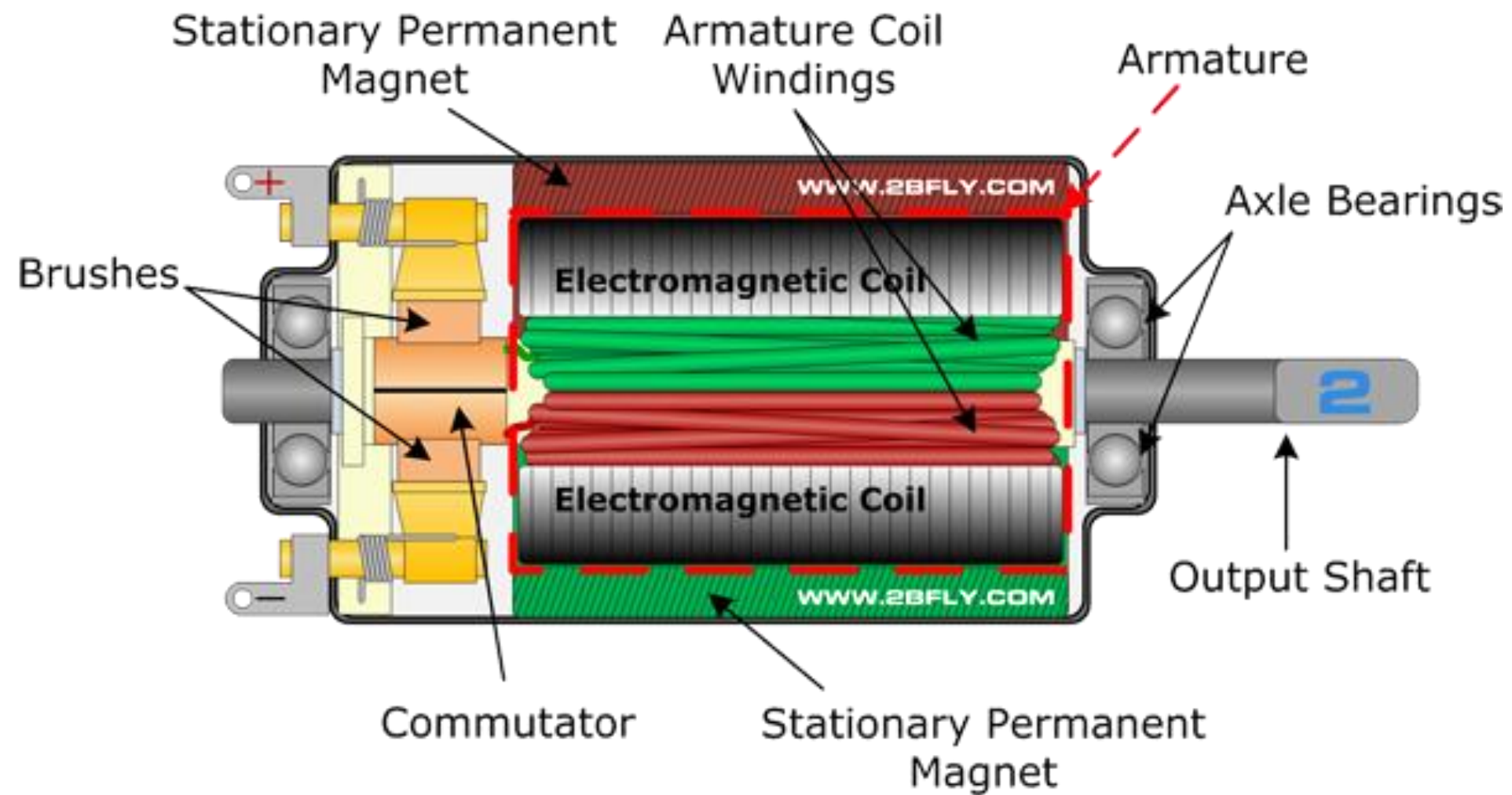
- Read a DC Motor data sheet and glean key information
- Formatted as Canvas quiz
- Grade is highest of two submissions

Direct Current (DC) Motors



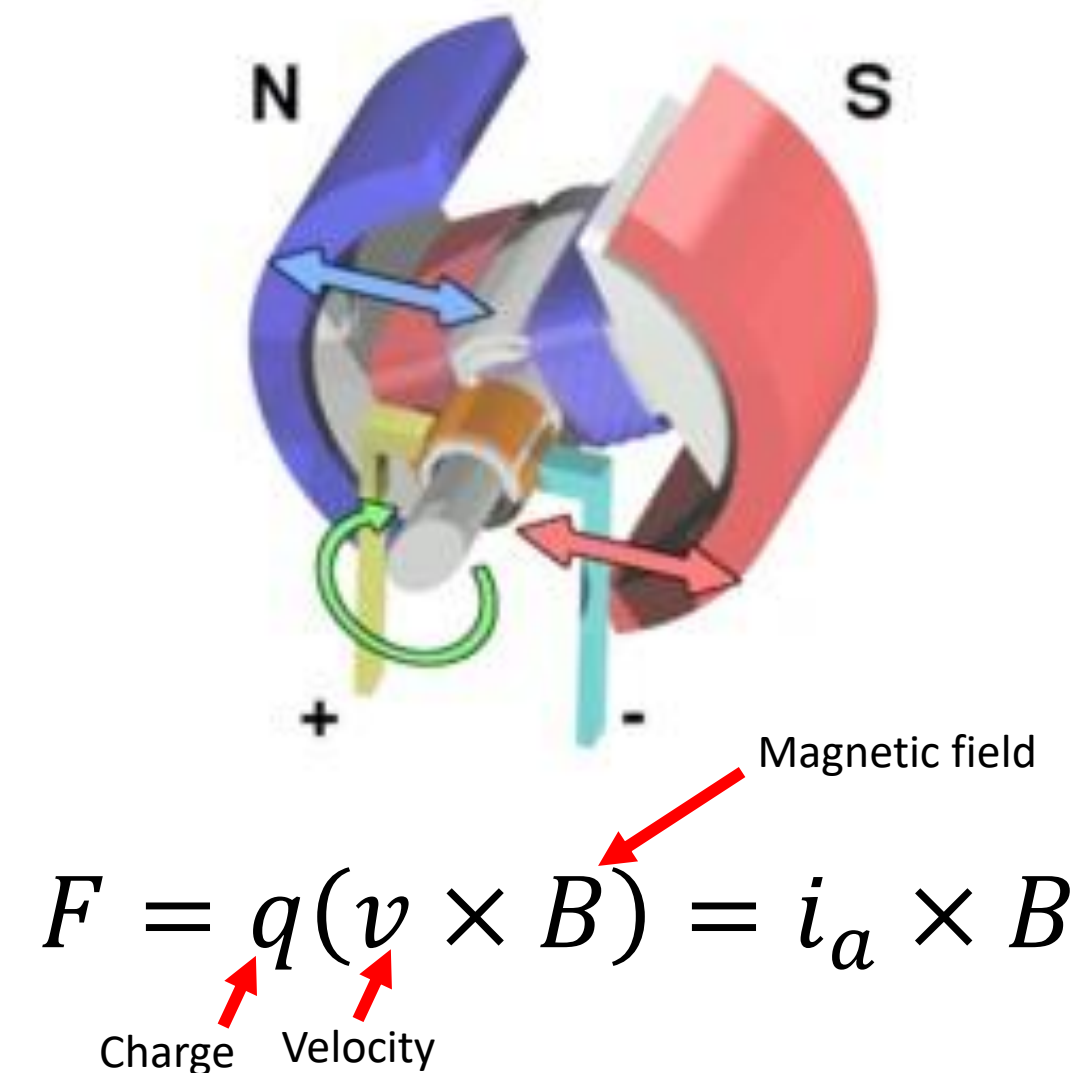
- Small, inexpensive, reasonably efficient, easy to use, ideal for small robotic applications
- Convert electrical energy into mechanical energy
- Limitations, including mechanical friction, cause some electrical energy to be wasted as heat
 - Toy motors: efficiencies of 50%, Industrial-grade motors: 90%+

Parts of a DC motor

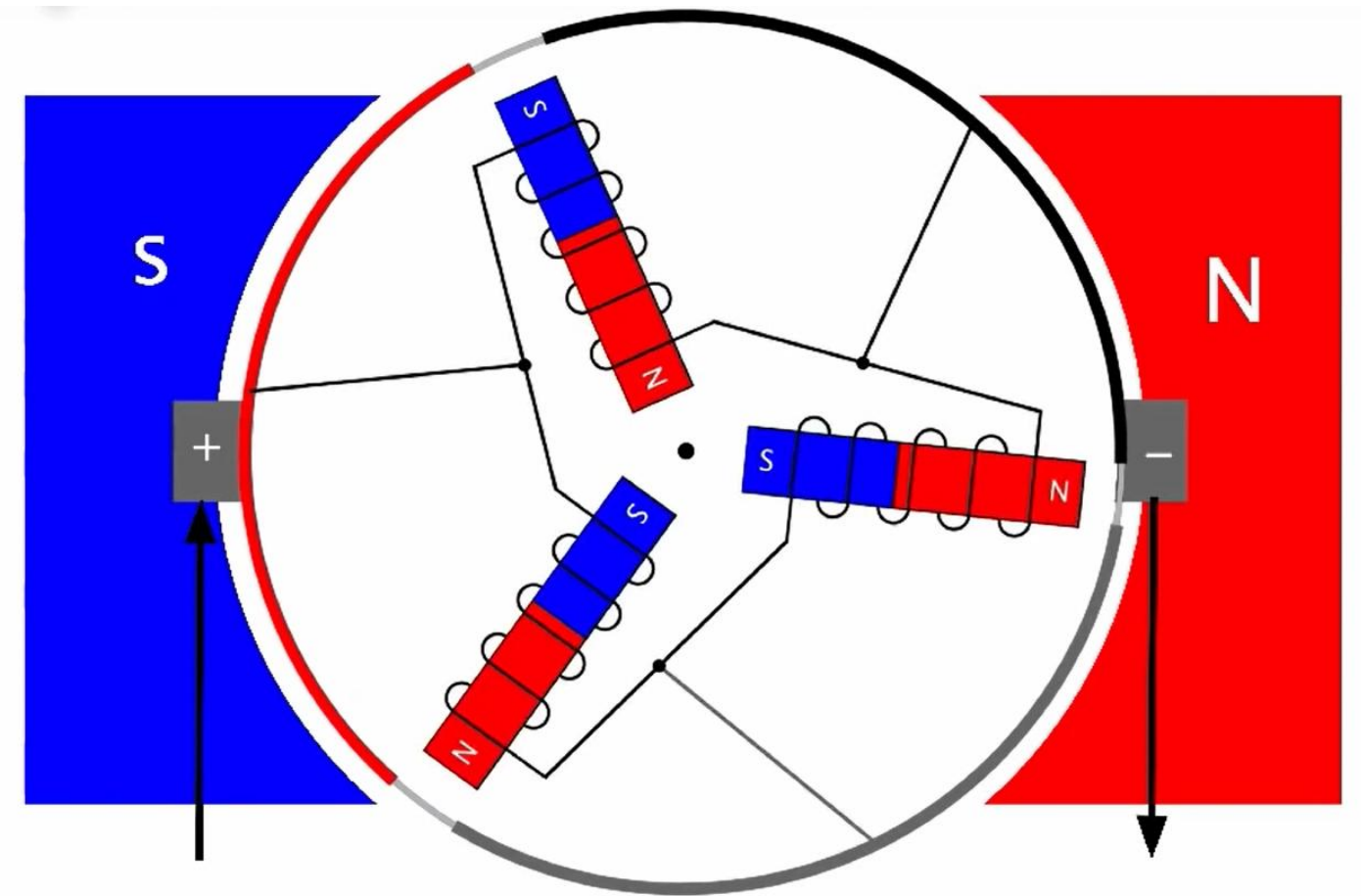
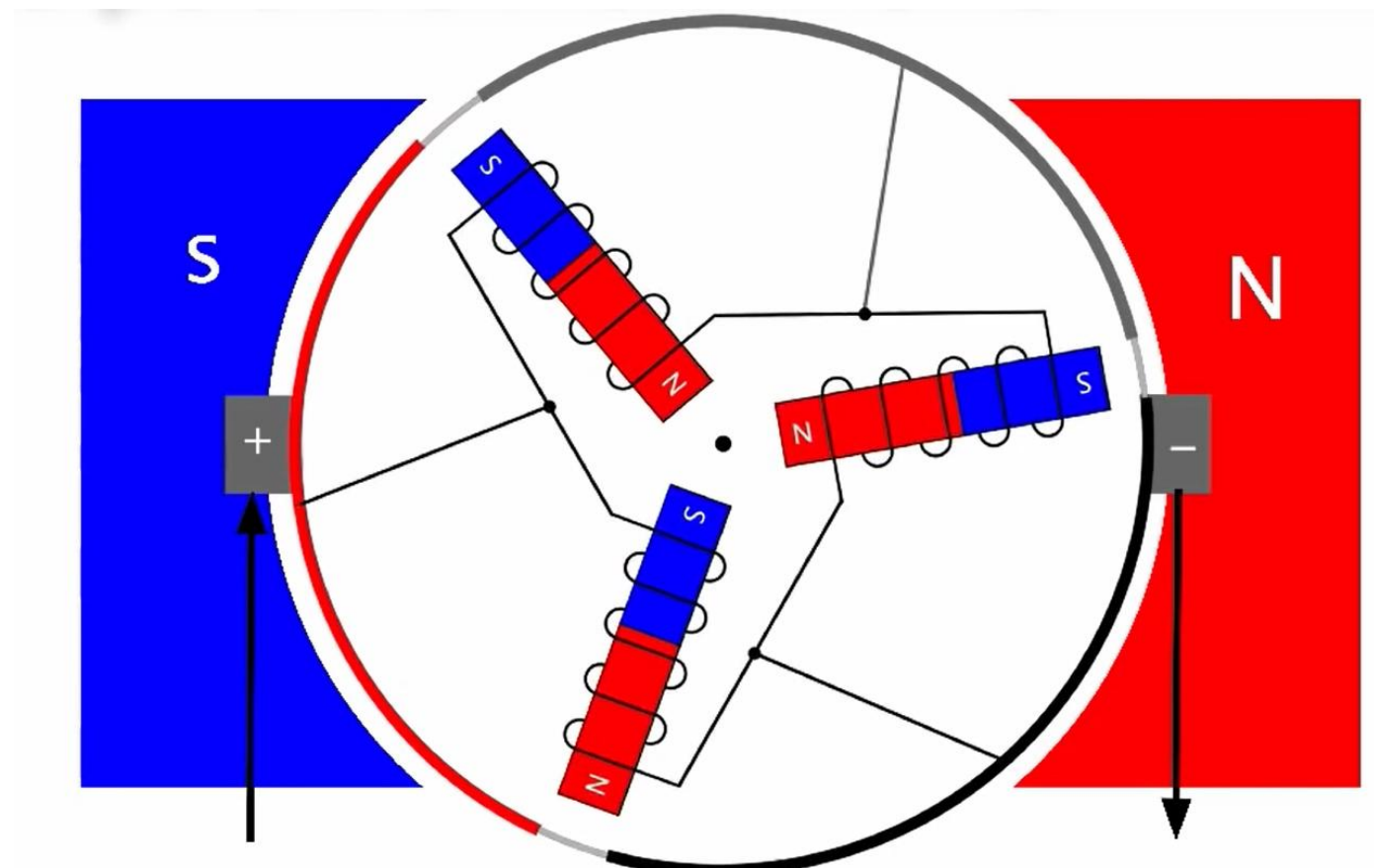


Brushed DC Motors – how do they work?

- By running electrical current through loops of wires mounted on rotating shaft (*armature*)
- When current is flowing, loops of **wire generate a magnetic field that reacts against the magnetic fields of permanent magnets** positioned around the wire loops
- These magnetic fields push against one another and the armature turns



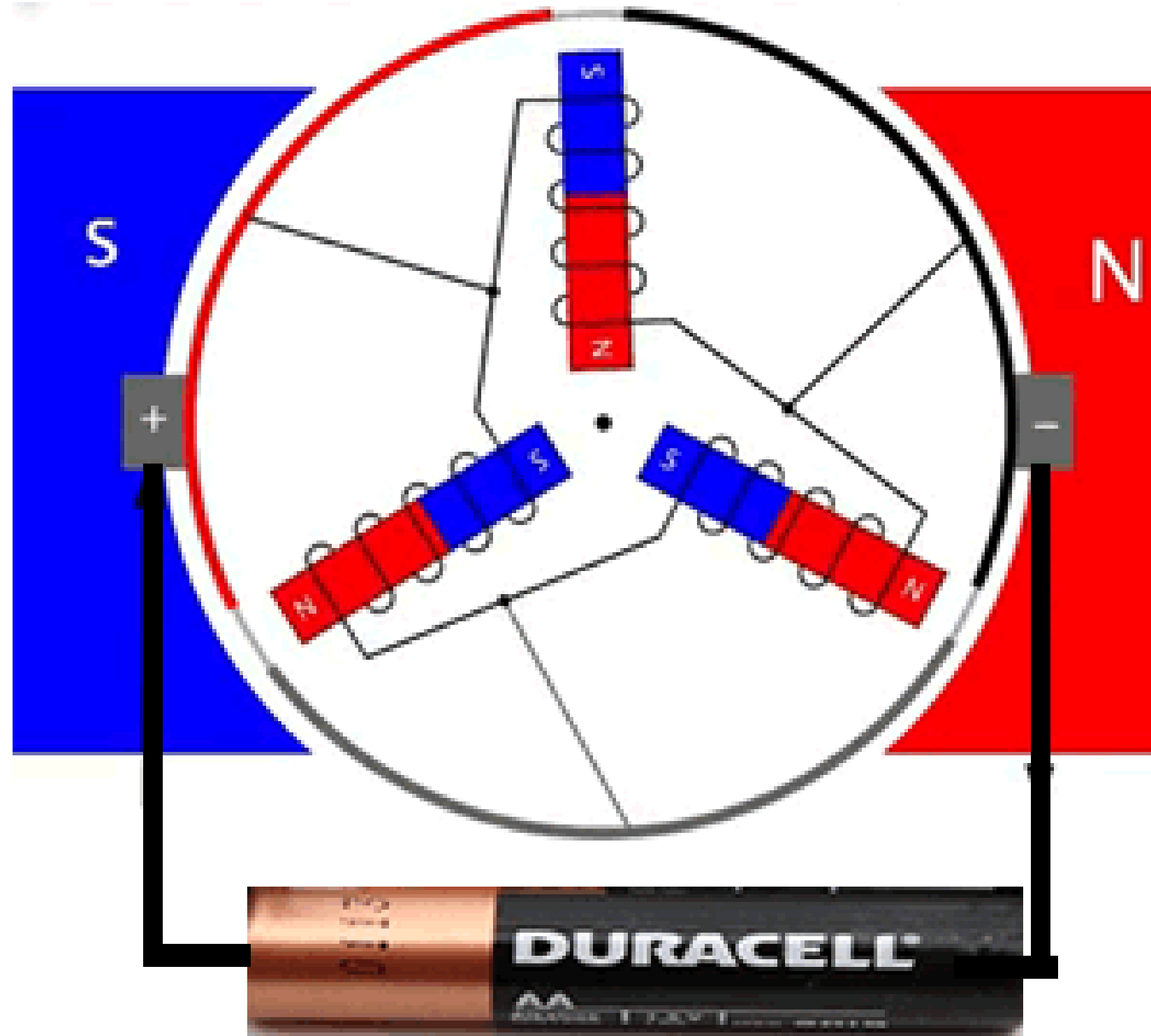
A 3-pole DC motor in action



Can we have more than 3-pole?

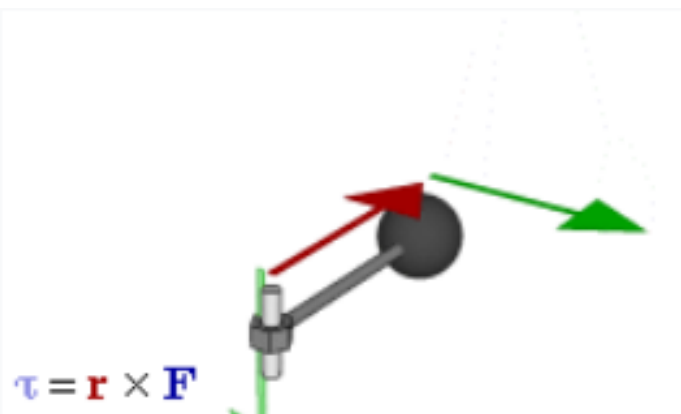
1: <http://www.dynamicscience.com.au/tester/solutions1/electric/3poledcmotot.html>
Youtube: <https://www.youtube.com/watch?v=1673-0Y3fFQ>

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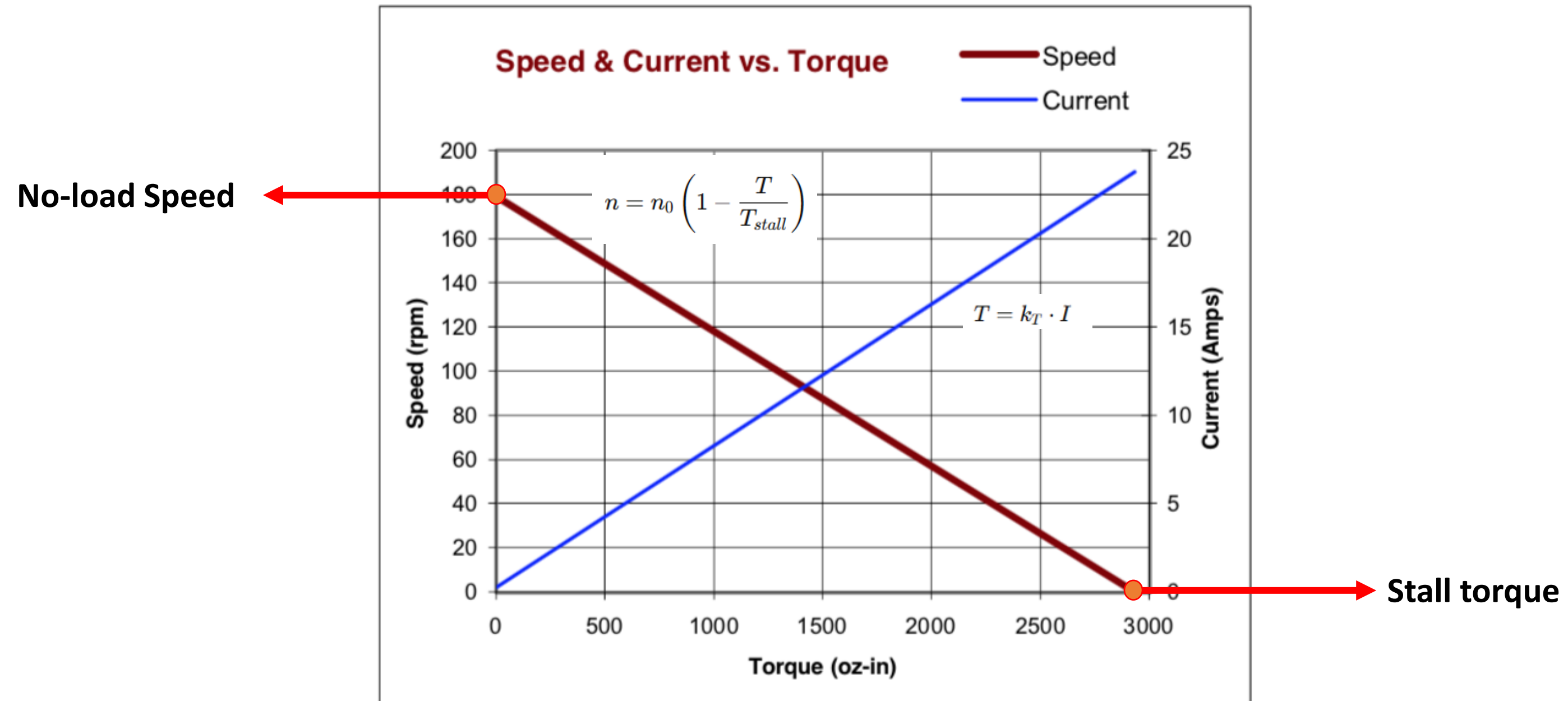
DC motor torque and power



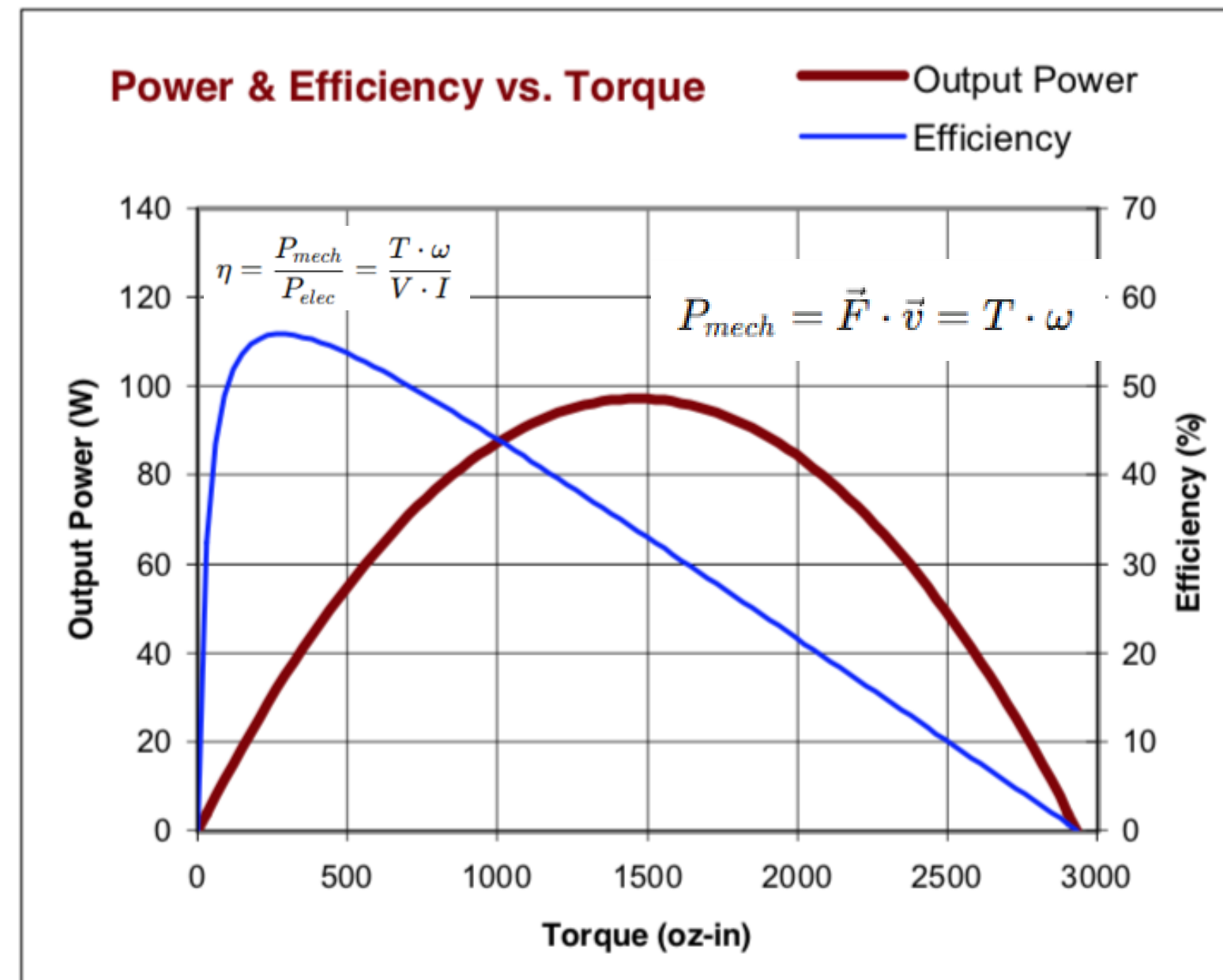
Category	Formula	Meaning of Symbols	
Mechanical torque	$T = \vec{r} \times \vec{F}$	T : torque (N·m); $\vec{r}$: position vector (m); $\vec{F}$: force vector (N).	
Mechanical power	$P_{mech} = \vec{F} \cdot \vec{v} = T \cdot \omega$	P_{mech} : mechanical power (W); $\vec{v}$: velocity vector (m/s); ω : angular velocity (rad/s).	
Electrical torque	$T = k_T \cdot I$	k_T : torque constant (N·m/A); I : current (A).	
Electrical power	$P_{elec} = V \cdot I$	P_{elec} : electrical input power (W); V : applied voltage (V).	
Efficiency	$\eta = \frac{P_{mech}}{P_{elec}} = \frac{T \cdot \omega}{V \cdot I}$	η : efficiency (0–1 or %).	

$$T_{mech} = T_{elec} - T_{loss}$$

Motor speed & current vs torque



Motor speed & power vs torque



is are shown for reference only.

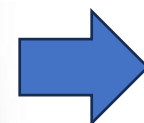
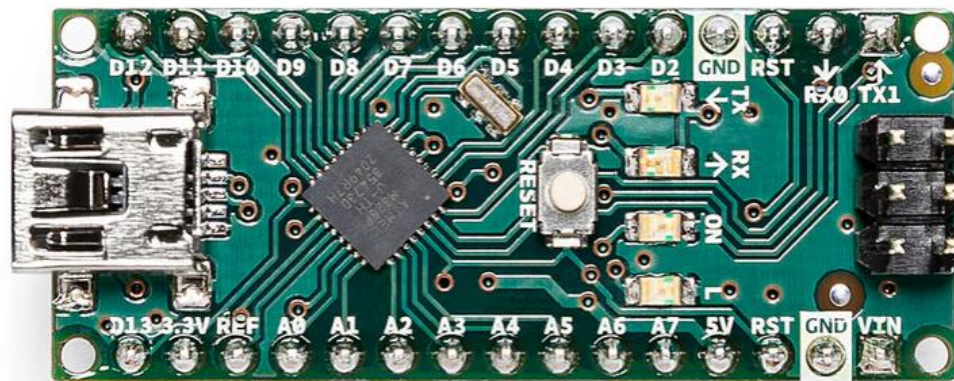
© 2001 Pittman.

Can you just hook an electric motor directly to an Arduino?

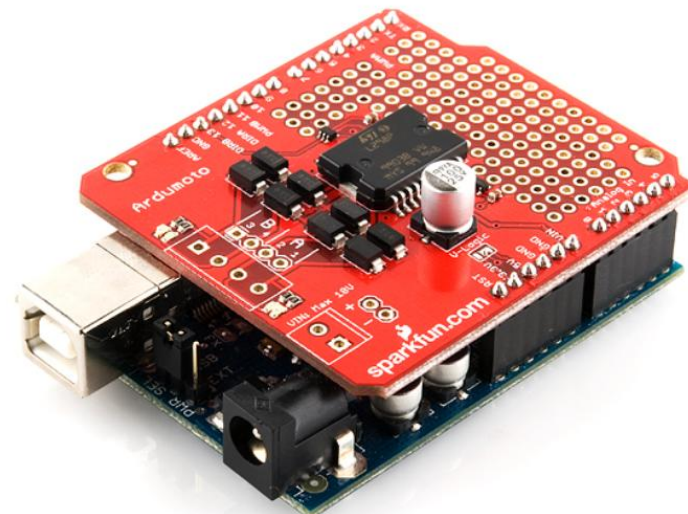
An Arduino's pins can only provide a **tiny amount of current (around 20–40 mA at 5 V)**. That's just enough to light an LED or read a sensor, not to power an electric motor. **Even very small DC motors draw hundreds of mA or more when running**, and several amps when starting or under load.

Connect motor to your Arduino can burn you Arduino Board!

Arduino: Brain, make decisions



Motor driver: Voltage switching, Current amplification, Direction control



Motor



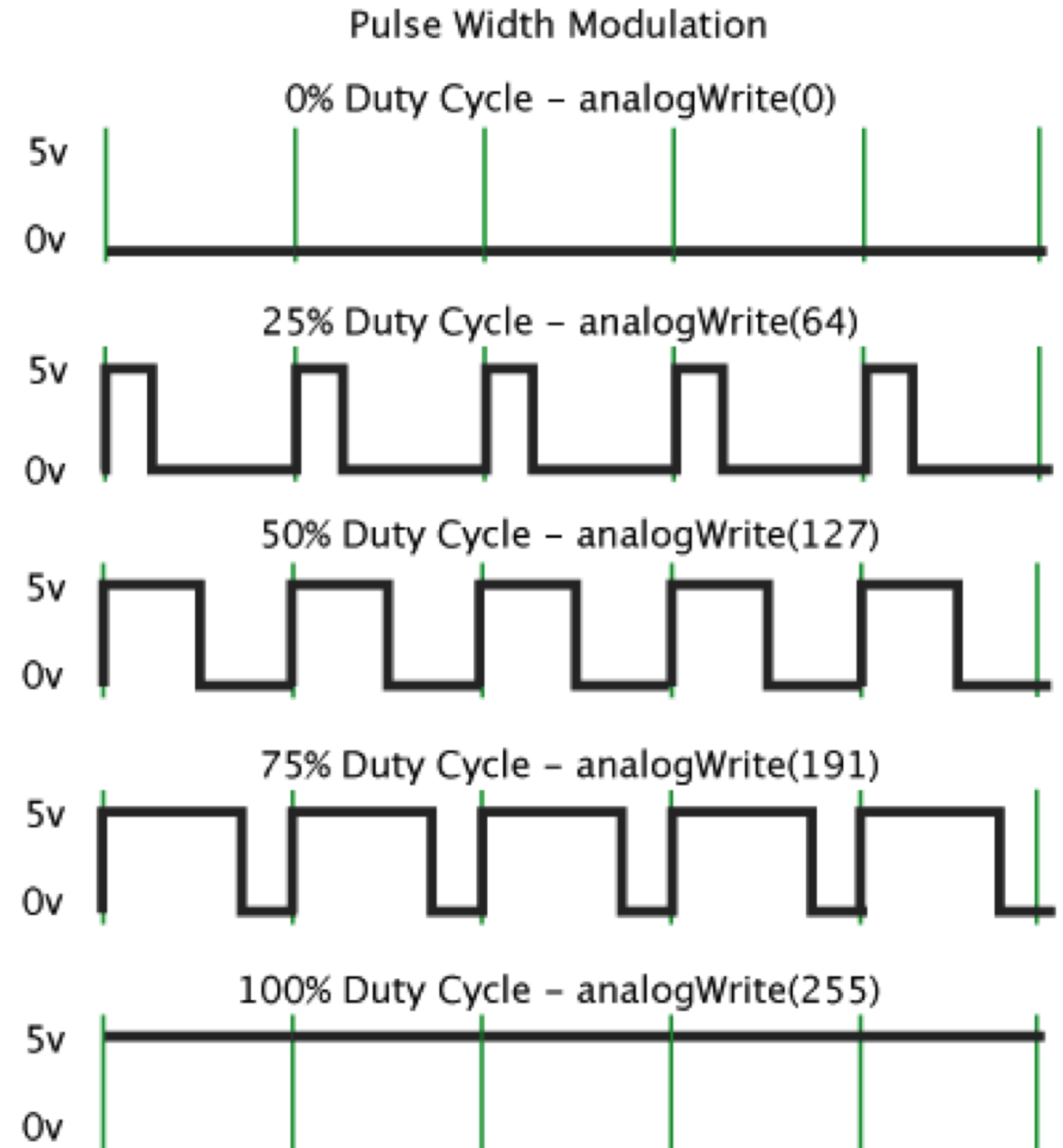
How do you control a motor?

Pulse-width modulation (PWM), change speed

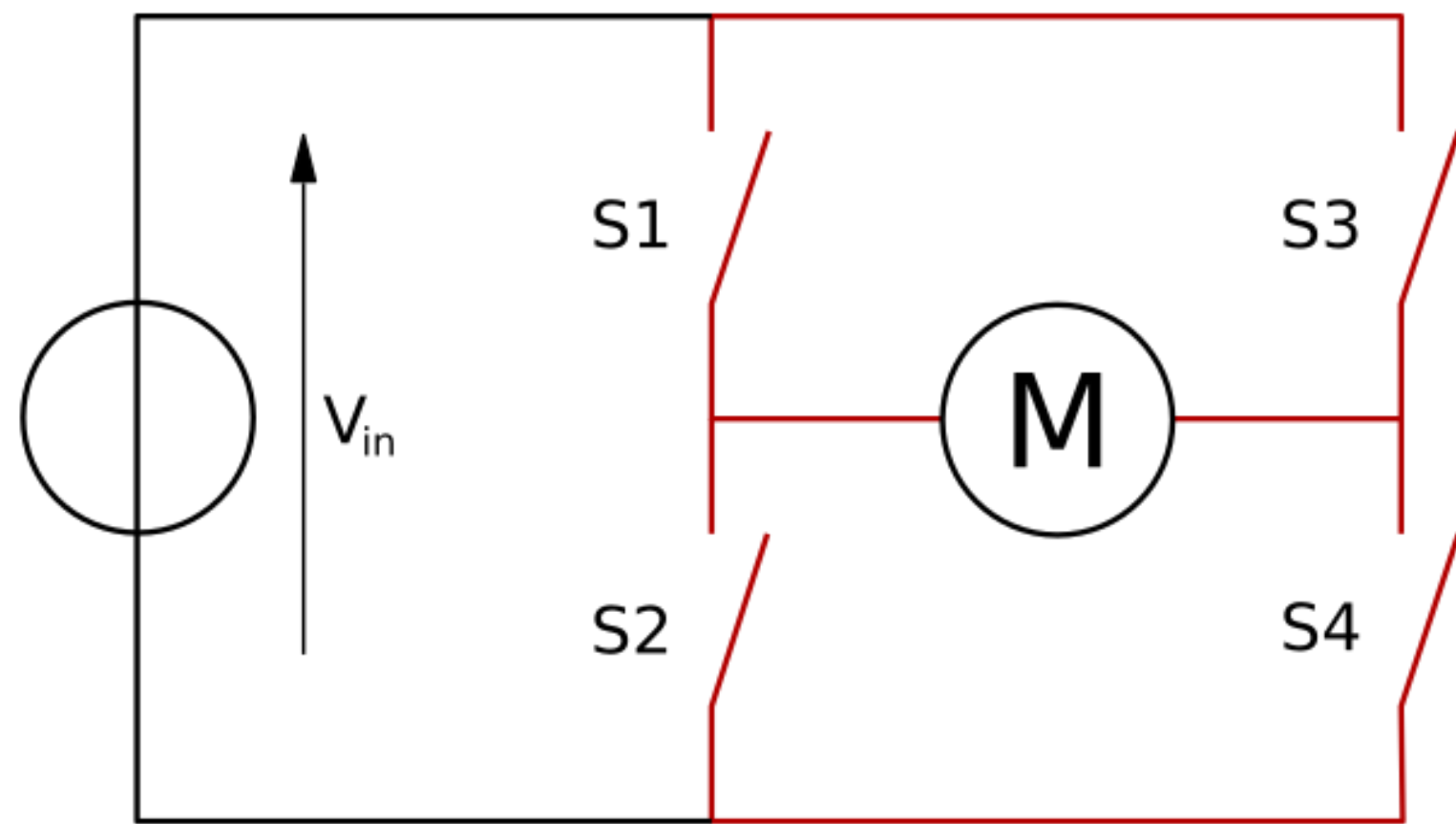
- Frequencies
 - Typically between 1-10 kHz
 - Arduino is about 490 Hz
- Duty Cycle
 - Percentage of time on in each period
 - On Arduino, 0-255 represents 0-100%
 - Use `analogWrite()`

$$V_{avg} = Duty\ Cycle \times V_{max}$$

$$\omega \approx \frac{V - I \cdot R}{k_E}$$

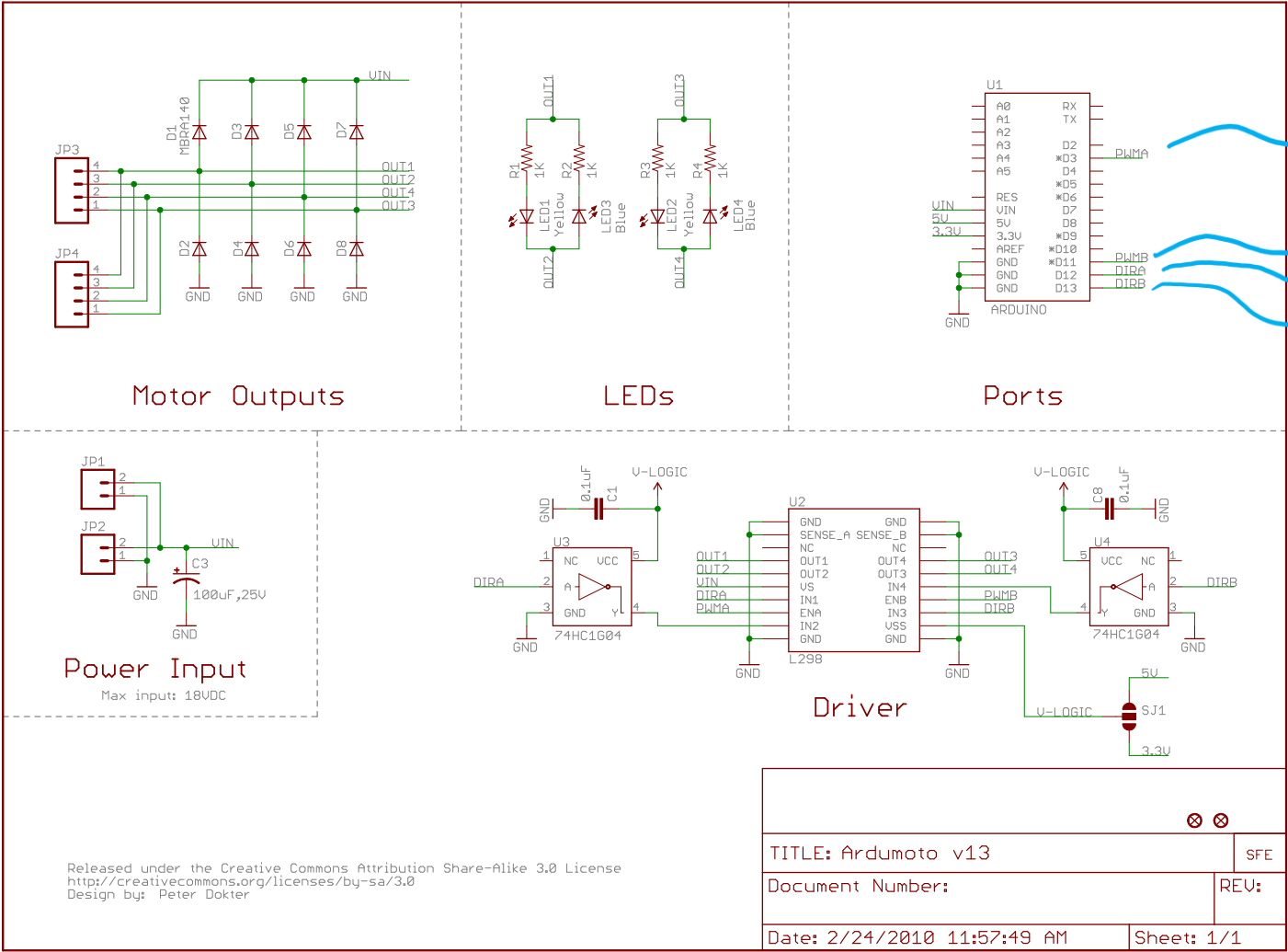
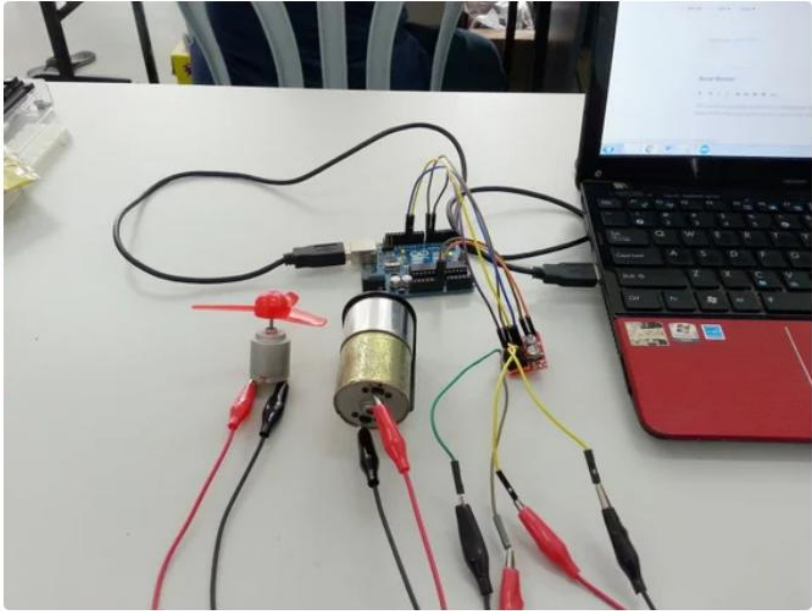
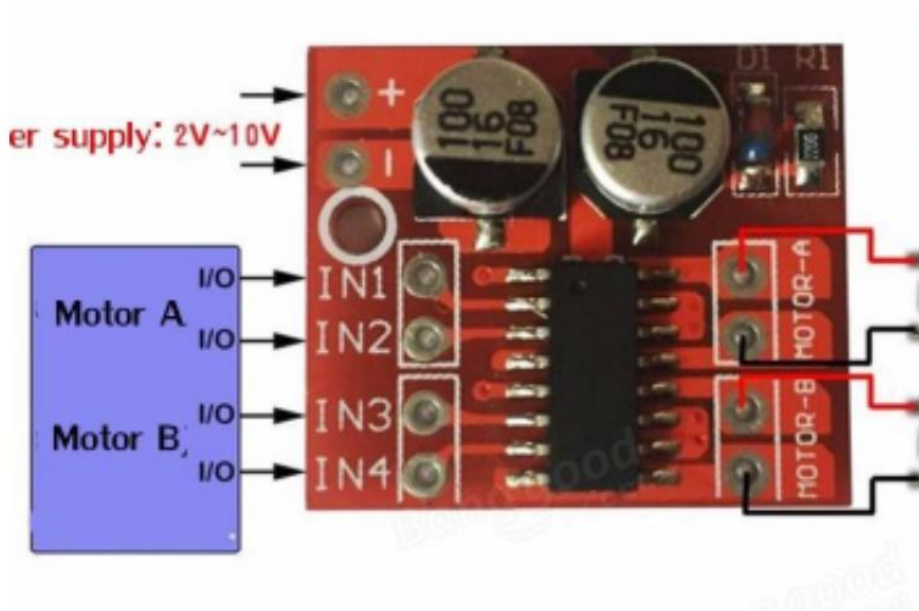


H-Bridges, change direction



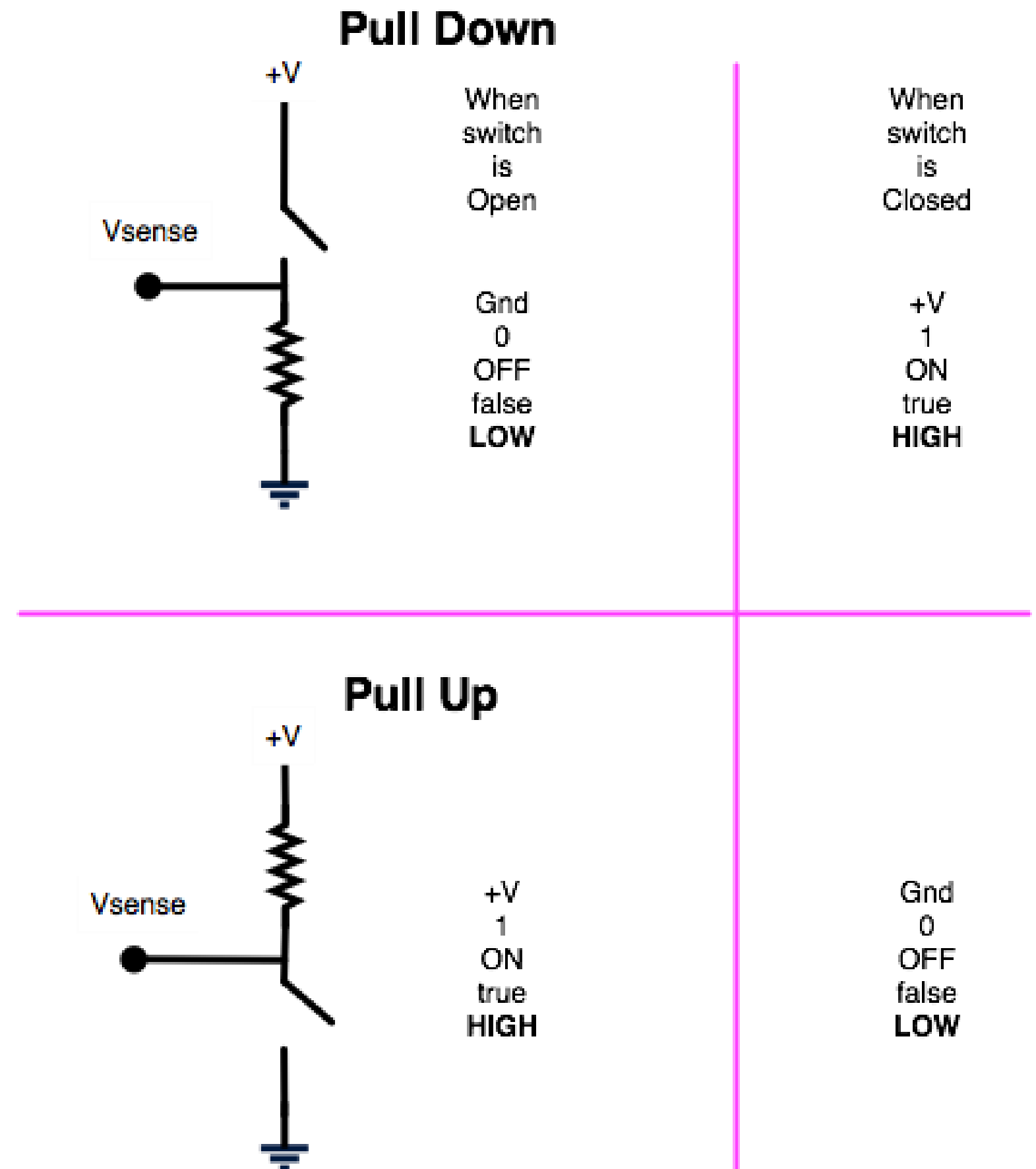
Switch State	Current Direction	Motor Effect
S1 + S4 closed	Left → Right	Forward
S2 + S3 closed	Right → Left	Reverse
All open	No current	Coast (free run, motor slows gradually)
S1 + S2 closed <i>or</i> S3 + S4 closed	Terminals shorted	Brake (fast stop)

Sparkfun Ardumotor Motor Driver Board



Wiring switches

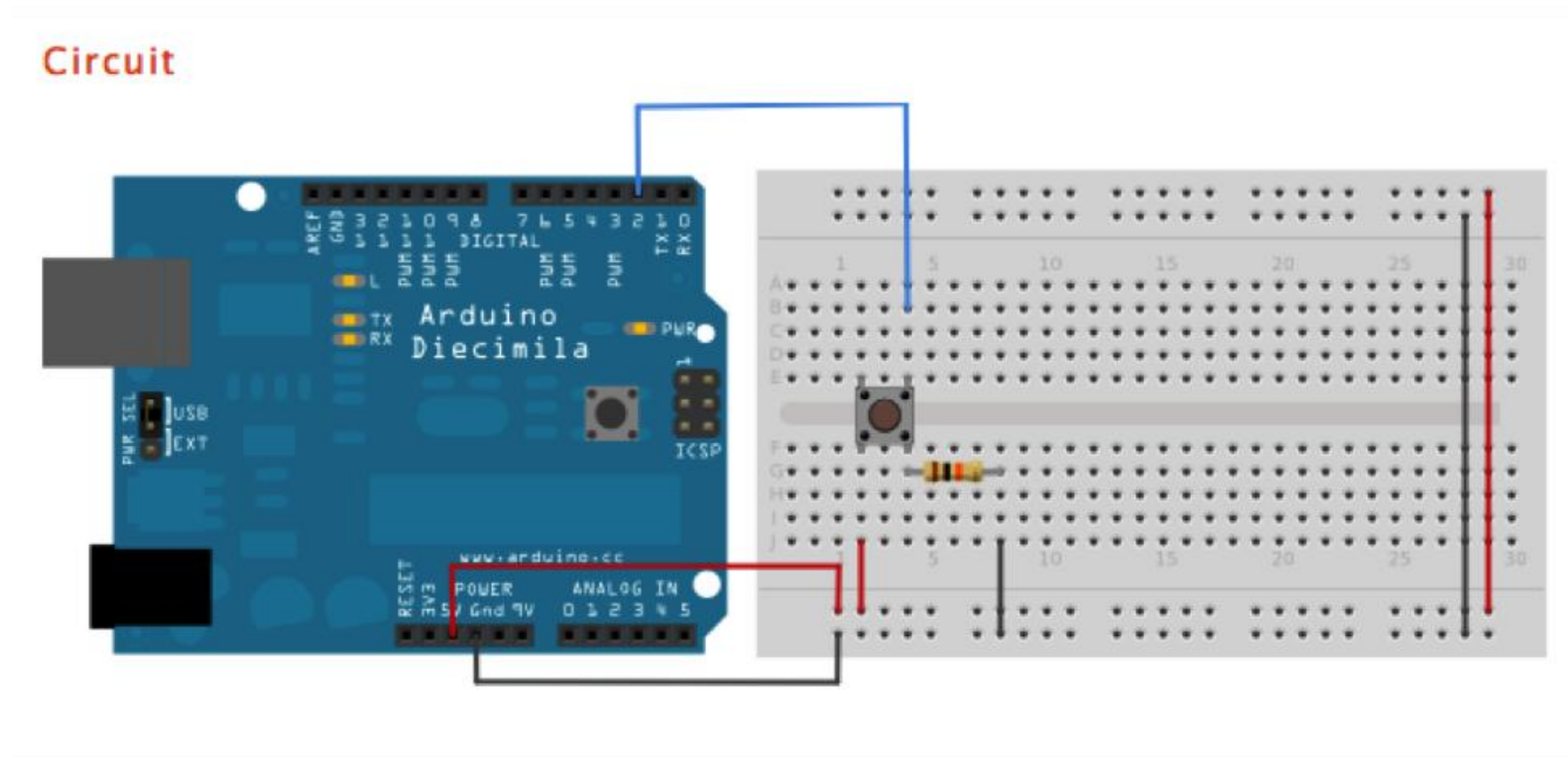
- Pull-Down Resistor
 - Logic works as expected
 - Needs an external resistor
 - `pinMode(pin, INPUT)`
- Pull-Up Resistor
 - Inverse logic
 - Resistor is internal to Arduino
 - `pinMode(pin, INPUT_PULLUP)`



Wiring switches

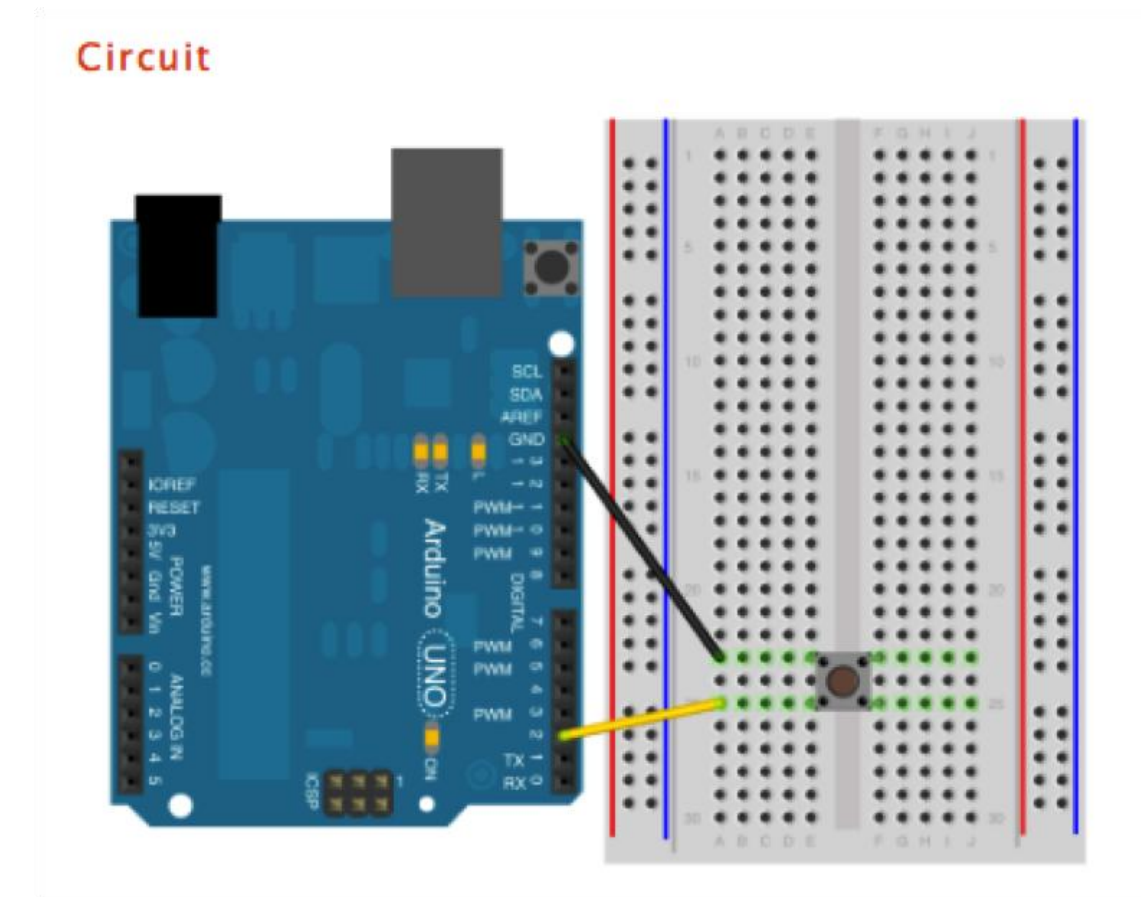
Pull-down

<http://arduino.cc/en/Tutorial/Button>

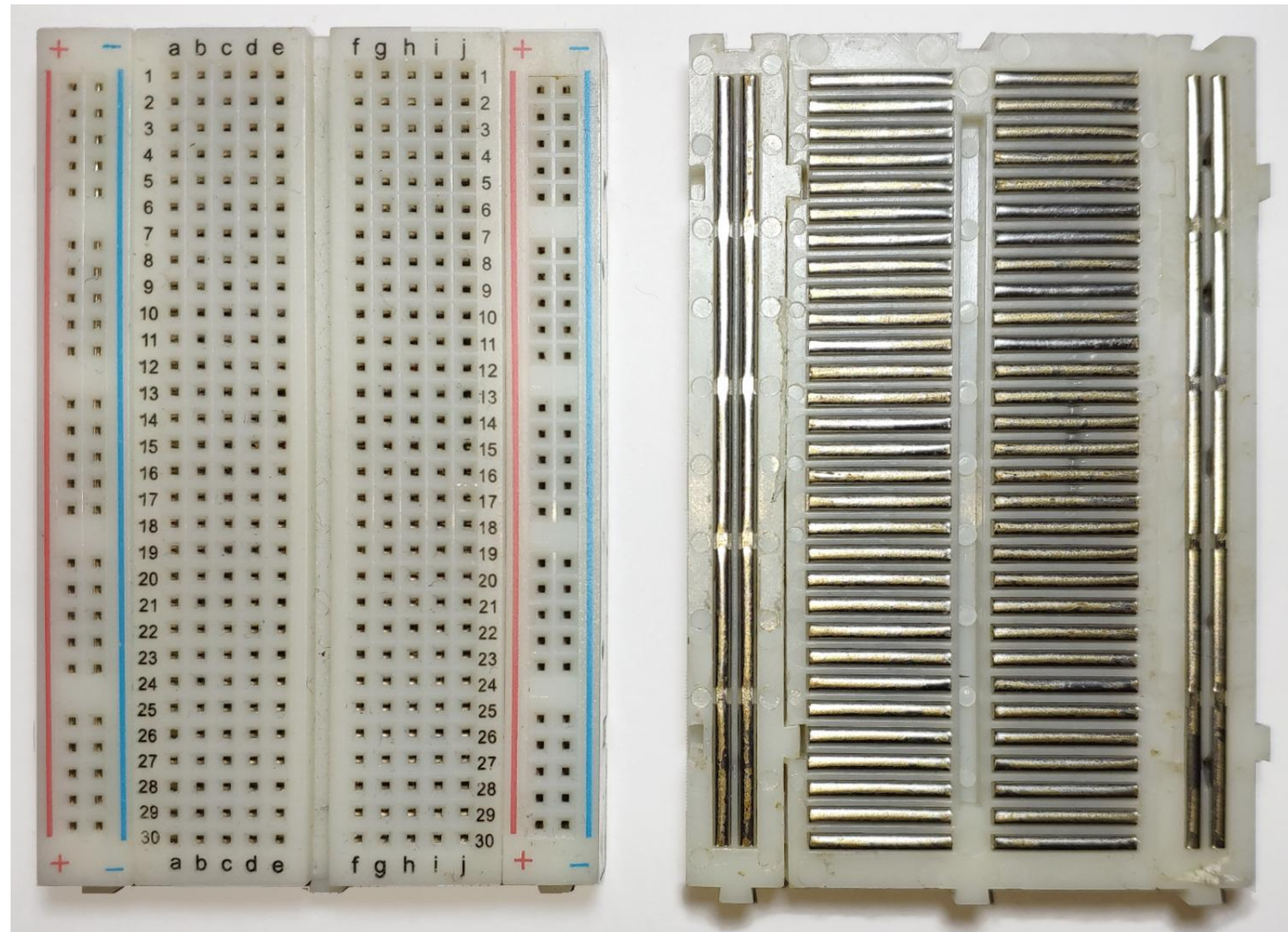


Pull-up

<http://arduino.cc/en/Tutorial/InputPullupSerial>



Breadboard



Lab 1 – Motor characterization

- Use your motors to lift a weight
- Measure average motor speed by timing the process of lifting
 - Generate speed, power plots
- Measure current to calculate electrical power
- Measure speed variation by varying pwm
- Measure reliability by repeating a trial
- Repeat for both motors